Sharbatanu Chatterjee, PhD

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শর্বতনু চ্যাটার্জি, বিদ্যাবাচস্পতি

Contact Information

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Research Interests

My general research interest is in systems neuroscience. I have experiences in a variety of methods - genetic tools, behavioural analysis, machine learning, and imaging. I am interested in using these approaches to discovering mechanisms by which networks of cells integrate internal states with external sensory input to produce robust yet flexible behaviour in animals (like fish, flies, rodents, humans, etc.).

Keywords : Systems Neuroscience, Behaviour, Computation, Imaging, Machine Learning.

Education

Sorbonne Université, France
PhD (Doctorat) in Physics, specialised in Neuroscience, Laboratoire Jean Perrin
Thesis supervisor : Dr. Volker Bormuth — *Dec 2020 - Dec 2024*

École polytechnique fédérale de Lausanne (EPFL), Switzerland
Master of Science, Life Sciences and Technology
Thesis supervisors : Prof. Carl Petersen & Prof. Wulfram Gerstner — *Sep 2016 - Aug 2018*

Indian Institute of Technology Kanpur (IIT Kanpur), India
Bachelor of Technology, Computer Science and Engineering
Minors in English Literature and Linguistic Theory — *Jul 2012 - Jul 2016*

Employment

Centre national de la recherche scientifique (CNRS)
Postdoctoral researcher in neuroscience (Chercheur en neurosciences)
Paris-Saclay Institute of Neuroscience (NeuroPSI) — *February 2025 - present*

Sorbonne Université
Research Engineer (Ingénieur de recherche) — *September 2020 - November 2020*
Laboratoire Jean Perrin — *December 2024 - January 2025*

Sainsbury Wellcome Centre, University College London (UCL)
Research Assistant,
Learning, Inference and Memory Lab, Dr. Athena Akrami — *January 2019 - August 2020*

Publications

- **S. Chatterjee***; G. Migault*; N. Beiza-Canelo*; G. Debrégeas; V. Bormuth, *"Directionally biased neuronal responses to pitch-axis vestibular stimulation in larval zebrafish compared to roll-axis responses"*, Communications Biology, 2026. DOI link.
- **S. Chatterjee***; N. Beiza-Canelo*; G. Debrégeas; V. Bormuth, *"Biomechanics and neural subtrates underlying posture control strategies in larval zebrafish"*, in preparation, 2026.
- **S. Chatterjee**; *"Stratégies comportementales & circuits neuronaux pour le contrôle postural (Behavioural strategies & neuronal circuits for postural control)"*, PhD thesis, 2024. DOI link
- Marius't Hart; ... Athena Akrami; ... **S. Chatterjee**; ... Diego Alzate-Correa et al. *"Neuromatch Academy: a 3-week, online summer school in computational neuroscience."*, Journal of Open SourceEducation, 5(49), 118. DOI link
- S. Bhattacharjee; **S. Chatterjee**; A. Banerjee; T. Y. Ho; K. Chakrabarty; B. Bhattacharya, *"Adaptation of Biochemical Protocols to Handle Technology-Change for Digital Microfluidics"*, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, 2016. DOI link

**equal contribution*

Posters & Talks

- **S. Chatterjee**; Timothee Sanchez; Tihana Jovanic, *"Neuromodulatory control of navigational strategies during taxis in Drosophila larvae"*, poster, NeuroFly, Leipzig 2026
- **S. Chatterjee**; Timothee Sanchez; Tihana Jovanic, *"Neuromodulatory internal state-dependent control of navigational strategies during taxis in Drosophila larvae"*, talk, DIF Anniversary Day, Paris 2026
- **S. Chatterjee***; N. Beiza-Canelo*; G. Debrégeas; V. Bormuth, *"Biomechanics and neural circuits underlying postural control strategies in larval zebrafish"*, poster, Computing Cords Workshop, Janelia Research Centre, Washington D.C., 2025
- **S. Chatterjee**; Timothee Sanchez; Tihana Jovanic, *"Effect of dopaminergic neurons in behavioral strategies of anemotaxis in Drosophila larvae"*, poster, Maggot Meeting, Leipzig 2025
- **S. Chatterjee** *"Biomechanics and neural circuits underlying postural control behavioural strategies in larval zebrafish"*, poster, Behaviour, Kolkata, 2025
- **S. Chatterjee** *"Distinct and Asymmetric Neuronal Responses to Pitch- and Roll-Axis Vestibular Stimulation in larval zebrafish"*, poster, FENS, Vienna, 2024
- **S. Chatterjee** *"Distinct and Asymmetric Neuronal Responses to Pitch- and Roll-Axis Vestibular Stimulation in larval zebrafish"*, poster, COSYNE, Lisbon, Portugal, 2024
- **S. Chatterjee** *"Behavioural strategies and neural circuits for postural control in larval zebrafish"*, talk, ZENITH Symposium, Paris, France, 2023
- **S. Chatterjee***; N. Beiza-Canelo*; H. Moulle; M. Ghosh; T. Panier; G. Debrégeas; V. Bormuth, *"Classification of brain states and behavioural motor response strategy to vestibular stimulation"*, poster, EDPIF La journée scientifique (Science day EDPIF), Paris, France, 2022
- **S. Chatterjee***; N. Beiza-Canelo*; M. Callachand; H. Moulle; M. Ghosh; M. Dommanget-Kott; T. Panier; G. Debrégeas; V. Bormuth, *"Behavioural strategies and brain-wide neural circuits driving postural control in larval zebrafish"*, poster, FENS Forum, Paris, France, 2022
- **S. Chatterjee***; N. Beiza-Canelo*; H. Moulle; G. Debrégeas; V. Bormuth, *"Strategies of postural control in larval zebrafish"*, poster, Imaging Structure and Function of the Zebrafish Brain Conference, Trondheim, Norway, 2022
  **equal contribution*

Previous Research Projects

Learning, Inference and Memory Laboratory, SWC, UCL
Learning of statistical structure of sensory inputs
Mentored by Athena Akrami *January 2019 - August 2020*

- The objective of this work is to investigate how animals can updates their sensory representations to optimise reward output
- I use a high-throughput semi-automated behavioural protocol to train animals (rats presently, and eventually mice) on a delayed match to sample auditory task
- I manipulate the statistical distribution of the sensory samples to test how this affects the priors formed in the animal's brain and how this is used in working memory and decision making
- I construct a (normative) model based on a Bayesian framework to analyse and predict the behaviour of the animal.

Laboratory of Sensory Processing, Brain Mind Institute, EPFL
Sensitivity to perturbation in computational models of the barrel column
Mentored by Carl Petersen and Wulfram Gerstner *March - August 2018*

- The objective of this work is to investigate computationally, the sensitivity to perturbation of network activity in a neuronal network model of a barrel column (C2) in the mouse primary somatosensory cortex.
- I use published data from the laboratory as well as optimise for other parameters not available to best fit the known properties of the biological tissues, yet generalise well enough.
- It turns out that the dynamics changes appreciably (measured by a defined sensitivity measure) to even the addition of a single spike in the network.
- I found out that one certain class of cells (L4 fast spiking neurons) are the most important factor in the runaway dynamics, and also investigated the important role of thalamic input in preserving spike timing consistency in the presence of perturbation.

Aging and Vision in Action Lab, Institut de la Vision, Paris, France
Calculating ego-motion, path integration and depth from optic flow
Mentored by Denis Sheynikhovich, under Angelo Arleo *May - August 2017*

- The objective of the work was to deal with estimating an observer's self-motion, depth perception as well as path integration from what the observer sees as changing intensity pattern on their retina (called optic flow).
- Several algorithms were tested and the simplest as well as most biologically plausible was selected and implemented to test the results in terms of optic flow processing.
- Based on the processing, hypotheses regarding the change of processing with age were suggested in tandem with experiments on optic flow on young and aged people.

Laboratory of Computational Neuroscience, Brain Mind Institute, EPFL
Sequences reproduce cortical bidirectional connectivity statistics
Mentored by Aditya Gilra and Marco Lehmann, under Wulfram Gerstner *March - May 2017*

- The objective of the work was to investigate the nature of information storage in neural networks.
- I simulated modified neural networks with connected single neurons and compared the statistics of their weights during storing attractors, a sequence of correlated patterns and finally a set of sequences of correlated patterns.
- We found out correlation in sequences do not affect the connectivity, but a set of correlated patterns reproduces connectivity statistics from experiments when it is optimised to store such sets. This might be thought of as a different 'type' of information being stored in such models.
- A report for the project can be found here.

Laboratory of Computational Neuroscience, Brain Mind Institute, EPFL
Multisensory integration with stochastic variational learning in spiking networks
Mentored by Aditya Gilra and Johanni Brea, under Wulfram Gerstner *May - August 2015*

- The objective of the work was to try and test the possibility of multisensory integration being feasibly modeled by recurrent spiking networks with more biologically plausible learning rules.
- Worked under my post-doctoral guides to implement the said model architecture based on Python.
- Developed an insight into the different steps and challenges of the modeling process.
- Acquired a fair understanding of the issues in designing and testing working models, apart from core issues in the physics and neuroscience behind the models.

Nanotechnology Research Triangle, Indian Statistical Institute, Kolkata, India
Digital Microfluidic Protocols
Mentored by Sukanta Bhattacharjee, under Bhargab B. Bhattacharya *May - August 2014*

- The objective of the work was to suggest new ways to deal with technology enhancement like processor speed increases in the context of digital microfluidic biochips (DMFB).
- Tried out various methods to avoid a complete re-synthesis of the given architecture (which would be a trivial solution), under the guidance of Sukanta Bhattacharjee
- Developed a method, based on symbolic encoding and SAT solvers, added with rich graph-theoretic procedures, to intelligently use the existing action sequences and save costly resources.
- Continued development on the project that culminated in the publication of the work in 2016.

Teaching Experience

- ***Biophysics***
  (Teaching Assistant) *2021, 2022, 2023*
  Responsible for teaching a group of Master's students of the Sorbonne Université about light-sheet microscopy, zebrafish neuroscience, and data analysis.
- ***Unsupervised and Reinforcement Learning in Neural Networks***
  (Teaching Assistant) *February - July 2018*
  Am responsible in helping students in the exercises of the course offered to master's students at

EPFL.

- ***Data Structures and Algorithms*** *(Teaching Assistant)* *August - November 2016*
  Was responsible for teaching, clearing doubts, preparing and grading questions as well as smoothly conducting the course organised by the Department of Computer Science and Engineering, IIT Kanpur.
- ***Introduction to Machine Learning*** *(Instructor)* *Summer 2016*
  Taught a course for a group of undergraduate and postgraduate students at IIT Kanpur, organised by the Association of Computing Activities. It was meant to introduce people to the field of machine learning, its practice and some of the theory behind it.

Skills

- ***Computational Tools:*** TensorFlow, PyTorch, Theano, Brian2, Neurolucida
- ***Bioinformatics Tools:*** BLAST, ClustalW2, HMMER, MEME suite
- ***Programming Languages:*** C, C++, Python, R, Java, Javascript, MATLAB, GNU Octave
- ***Natural Languages:*** Bengali (native), Hindi/Urdu (proficient), English (proficient), Odia, French (proficient)

Course Projects

Visualising Cancer Data
Mentored by Kirell Benzi, EPFL *September 2017 - present*

- Used visualisation techniques to represent cancer genomic data on an interactive graph using the information from The Cancer Genomic Atlas, with clusters based on a novel distance measure. The method was developed by Rachel Jeitziner at the Brisken Lab of EPFL.
- Presented a poster on it at Google Research Day 2018 at EPFL.
- A dedicated website on the project may be accessed here. It is presently being improved.

Predicting the probability of Parkinson's disease from fMRI data
Mentored by Robert West, EPFL *September 2017 - January 2018*

- Used a variety of techniques on the PPMI data (Parkinson's Progression Marker's Initiative) which consists of resting state fMRI of controls and Parkinson's patients, to classify them into diseased and non-diseased patients
- Received best presentation at Applied Machine Learning Days 2018 conference, Lausanne, Switzerland.
- A dedicated website on the project may be accessed here. It is presently being improved.

Predicting the movement of a robotic arm via neural spikes
Mentored by Ricardo Chavarriagga, EPFL *September - November 2016*

- Used a variety of machine learning techniques to reduce the dimension of the data, sort spikes, and apply regression analysis to predict the motion of a robotic arm controlled by a monkey.

Learning fast kernels
Mentored by Harish Karnick, IIT Kanpur *September - November 2015*

- Acquired a fair understanding of the prevalent issues and techniques in learning of fast kernels, useful in ubiquitous kernel learning methods. Implemented the Fastfood based approach and proposed a possible technique, based on the À la carte algorithm.

Billion word imputation
Mentored by Harish Karnick, IIT Kanpur *January - April 2015*

- Implemented an approach based on N–grams based to predict missing words in English language sentences. This was followed by implementing a recurrent neural network (RNN) based approach (T. Mikolov *et al.* 2010) that accounts for context generated by the preceding part of the sentence while predicting the missing word, by using the network output as an input to the neural network.

Hallucinations in Garden Path sentences -
Mentored by Amitabha Mukerjee, IIT Kanpur *August - November 2014*

- Analyzed the processing of garden path sentences, true and hallucinated, through empirical data and designed self-paced reading based experiments as well as gaze-tracking experiments for the same. Tested hypotheses in the field (R. Levy *et al* 2011) of sentence-processing through human cognition models using collected data.

– Awarded seconnd-best course project for the semester.

Linguistic profiles of the Lamkang language
Mentored by Chaithra Puttaswamy, IIT Kanpur *August - November 2014*

– Carried out a detailed study of the grammar, vocabulary, phonology and morphology of the Lamkang language, spoken in Manipur in north-east India. Identified and discussed points of contact and convergence with local Meitei and Bishnupriya languages, as well as with Indo-Aryan languages like Bengali and Assamese.
– Awarded an A*, given only for exceptional performance.

RHadoop Server
Mentored by Arnab Bhattacharya, IIT Kanpur *August - November 2014*

– Aim of the project was to set up a Hadoop Cluster on IBM Blade server and configure RHadoop packages on it to allow distributed computing of R map reduce programs on this cluster.
– Understood and made usable the IBM blade server which was an unused resource in the CSE Department for the past 5 years. Selected as the best course group project among 30 others.

Positions of Responsibility

- **Academic Delegate, SV MA4, AgePoly, EPFL**, working as a link between the student body and the governing body of EPFL.*(February 2018-July 2018).*
- **Student Nominee, Department Undergraduate Committee**, working as a link between the student body and the governing body of IIT Kanpur. *(2014-15)*
- **Academic Mentor** with the Counselling Service, IIT Kanpur during the academic year 2013-14, to guide juniors and classmates in their academics, particularly for the courses **Mathematics-I**, **Introduction to Electrodynamics** and **Introduction to Biology**. *(2013-14)*
- **Coordinator, NERD (Notes on Engineering, Research and Design), IIT Kanpur**, responsible for organizing lectures on popular science and science communication, publishing and editing the student research magazine on science and technology and keeping the campus updated on latest news on science and technology, as a part of the NERD team. *(2013-15).*
- **Co-Editor, Vox Populi, IIT Kanpur**, responsible for maintaining and editing the campus newspaper as a part of a vibrant team. *(2014-15).*

Scholastic Achievements

- Recipient of the **Fin de thèse award** from the Fondation de la Recherche Medicale of France (2023-2024)
- Recipient of the **MSC Actions ITN fellowship** for my PhD, awarded by the European Commission (2020-2023)
- Two time recipient of the **Certificate of Merit for Academic Excellence**, awarded by IIT Kanpur for the academic years 2012-13 and 2013-14.
- Awarded the **Ram Parkash Chopra Memorial Scholarship** for the academic years 2014-15 and 2015-16, given to students for academic performance and aptitude.
- Awarded the **KVPY (Kishore Vaigyanik Protsahan Yojna) Fellowship** for students interested in basic sciences for 2011 and 2012 by the Government of India.